

A. Independent cross-sectional inputs

The starting point is two independent cross-sectional samples: prior-grade reference percentiles $U = F_X^{\text{ref}}(x_i)$ and current-grade reference percentiles $V = F_Y^{\text{ref}}(y_j)$.

Crucially, there is *no student-level linkage* between these samples—we cannot match any u_i to its corresponding v_j . This is exactly the situation faced by international assessments such as TIMSS, where Grade 4 and Grade 8 cohorts are sampled independently.

The distributions of U and V are the only observables; all subsequent inference proceeds from these marginals alone.